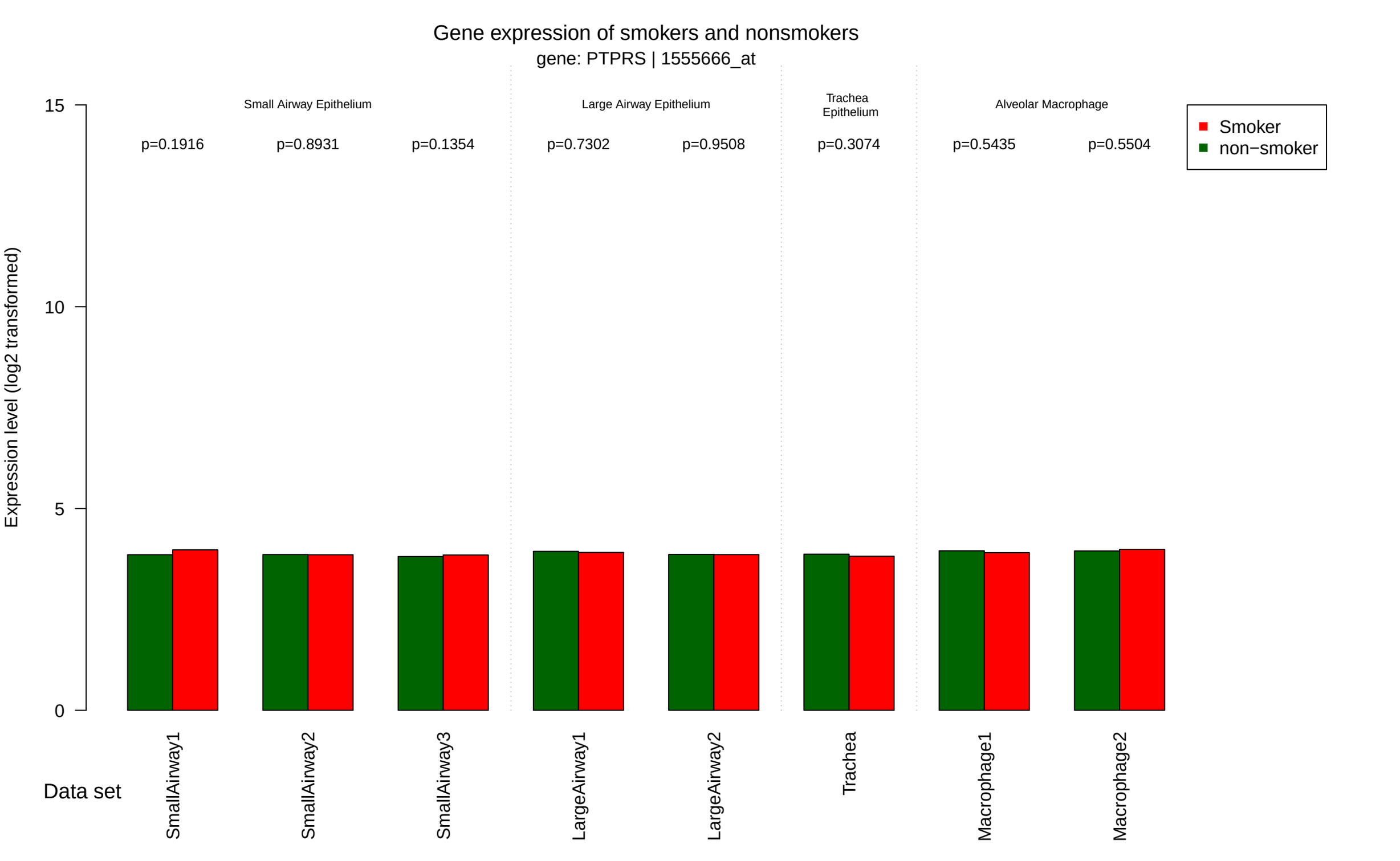
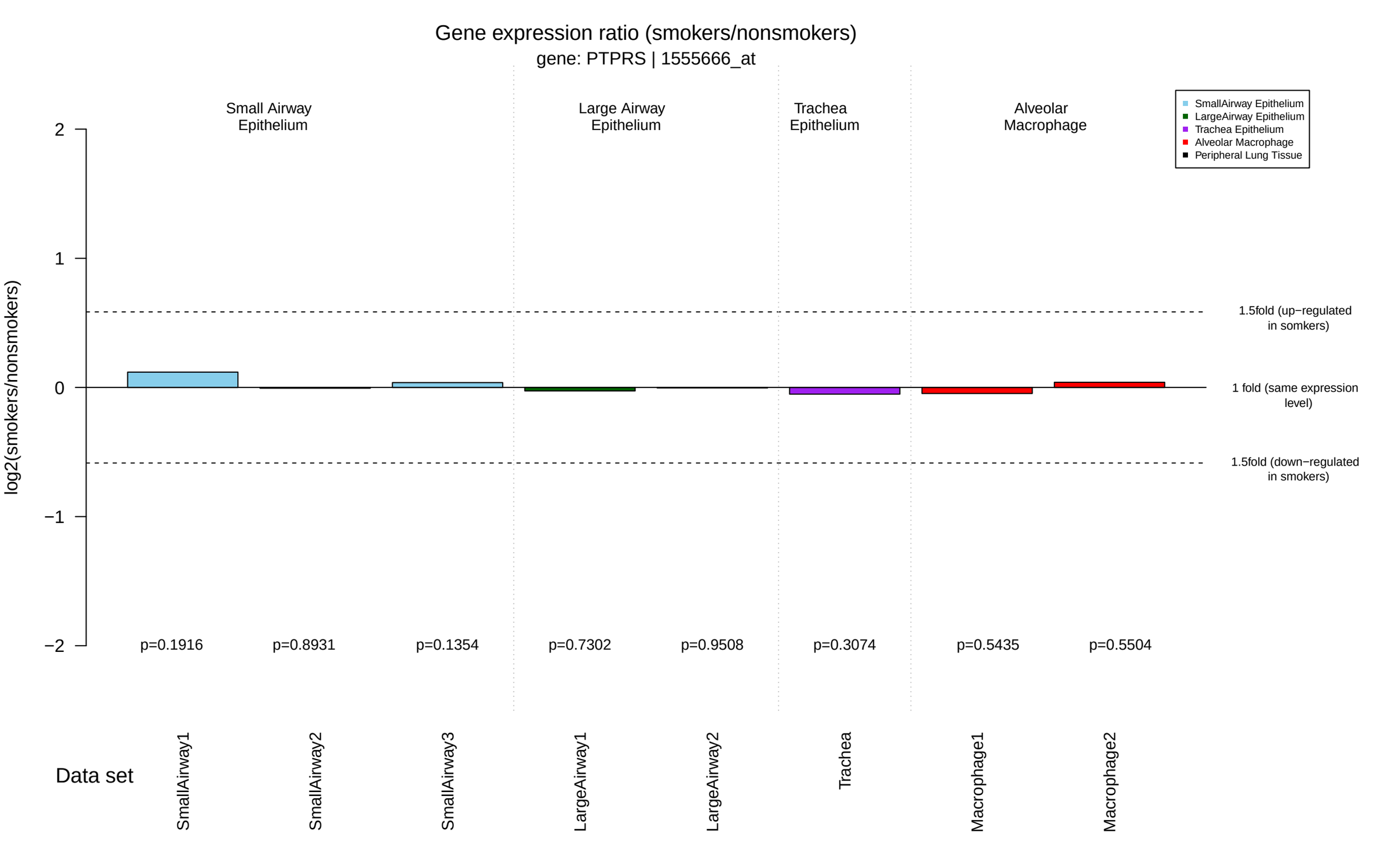
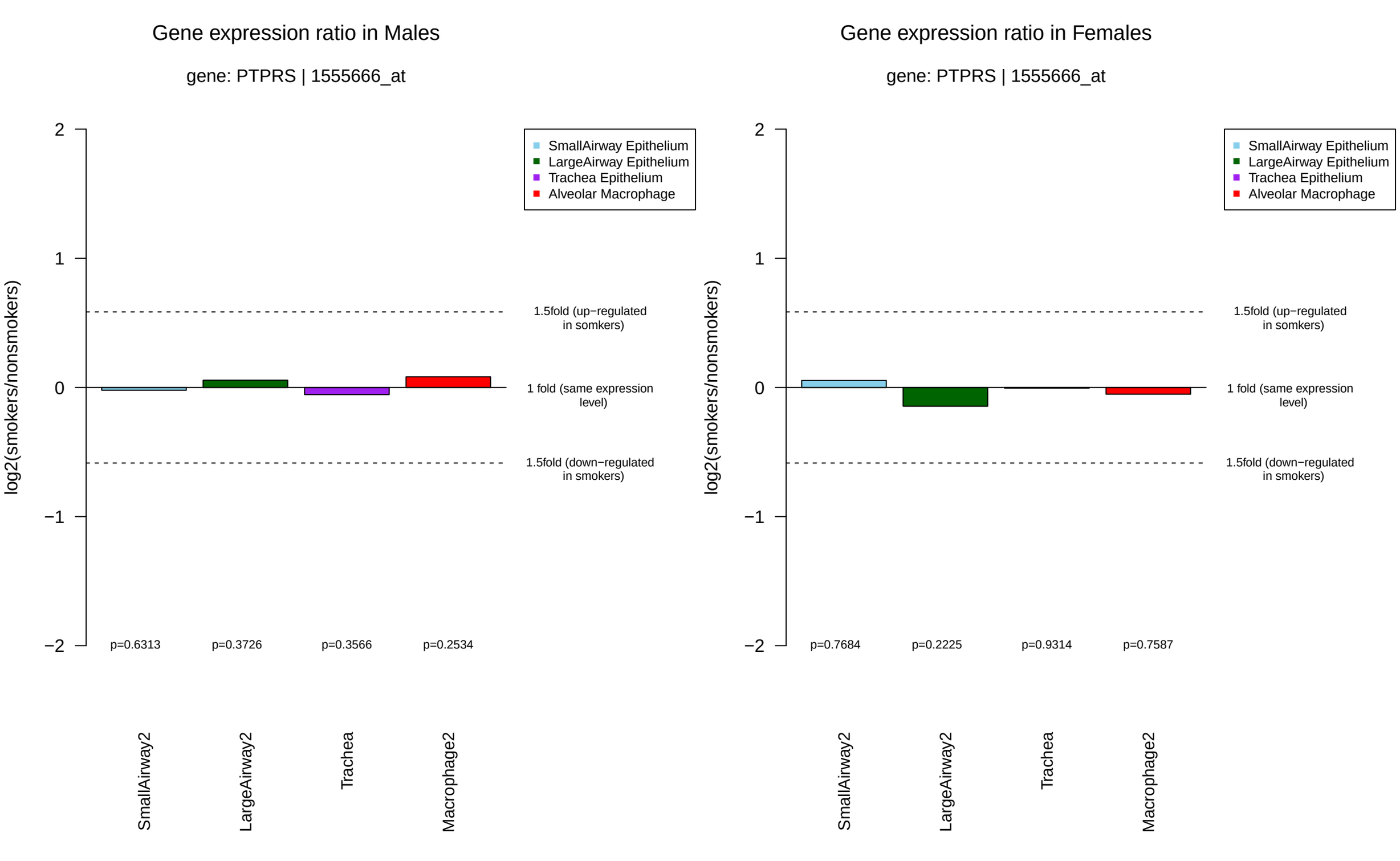
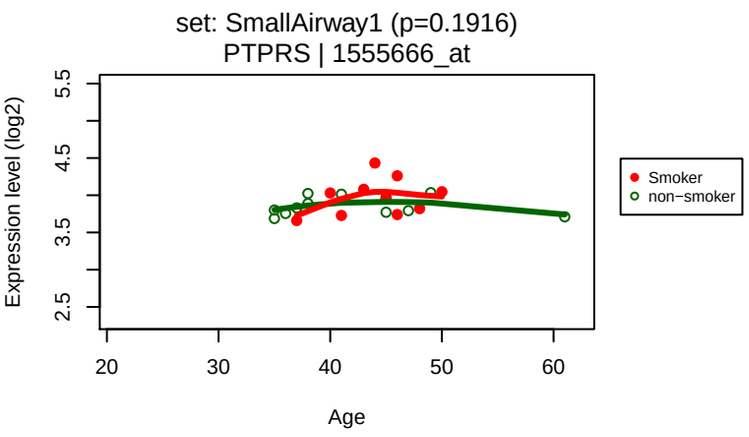
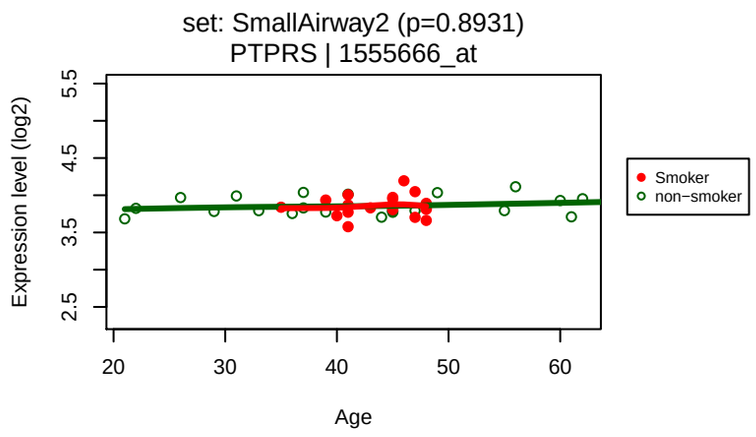
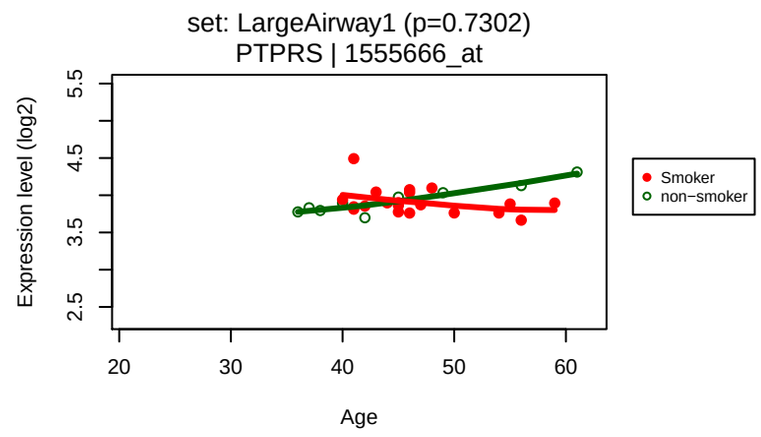

Gene expression of smokers and nonsmokers
gene: PTPRS | 1555666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1916
p=0.8931
p=0.1354
p=0.7302
p=0.9508
p=0.3074
p=0.5435
p=0.5504
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

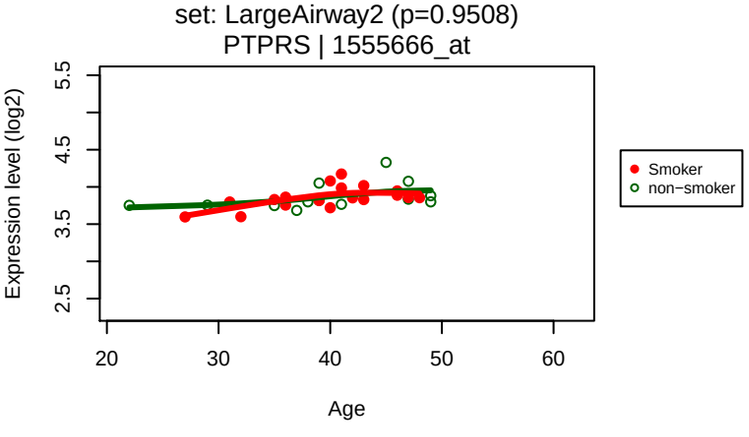

Gene expression ratio (smokers/nonsmokers)
gene: PTPRS | 1555666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1916
p=0.8931
p=0.1354
p=0.7302
p=0.9508
p=0.3074
p=0.5435
p=0.5504
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

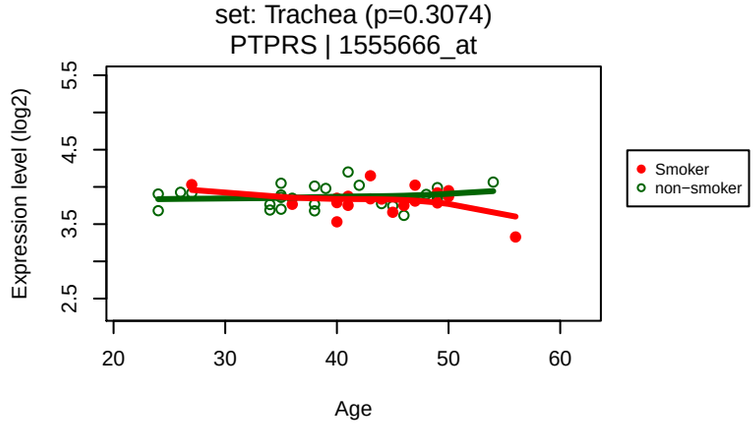

Gene expression ratio in Males
gene: PTPRS | 1555666_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6313
p=0.3726
p=0.3566
p=0.2534
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPRS | 1555666_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7684
p=0.2225
p=0.9314
p=0.7587
SmallAirway2
LargeAirway2
Trachea
Macrophage2

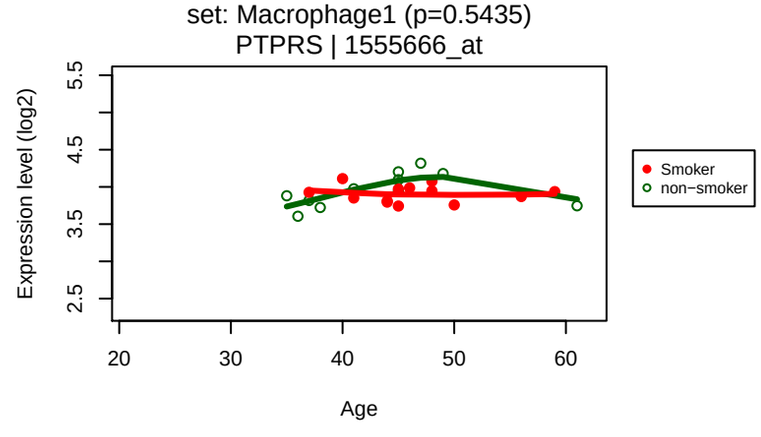

set: SmallAirway1 (p=0.1916)
PTPRS | 1555666_at
Expression level (log2)
Age
Smoker
non−smoker

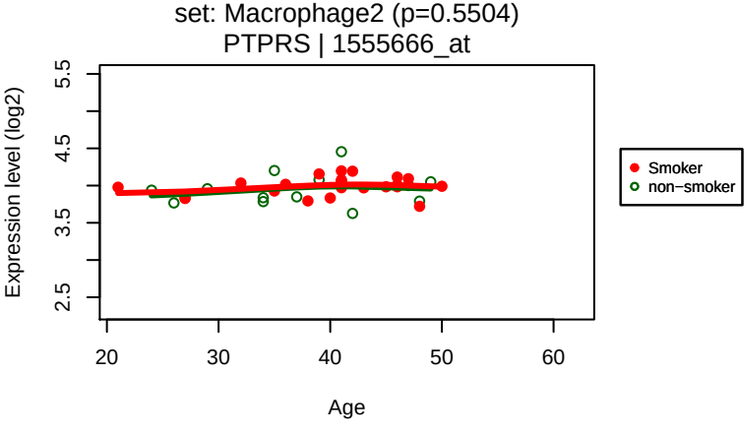

set: SmallAirway2 (p=0.8931)
PTPRS | 1555666_at
Expression level (log2)
Age
Smoker
non−smoker

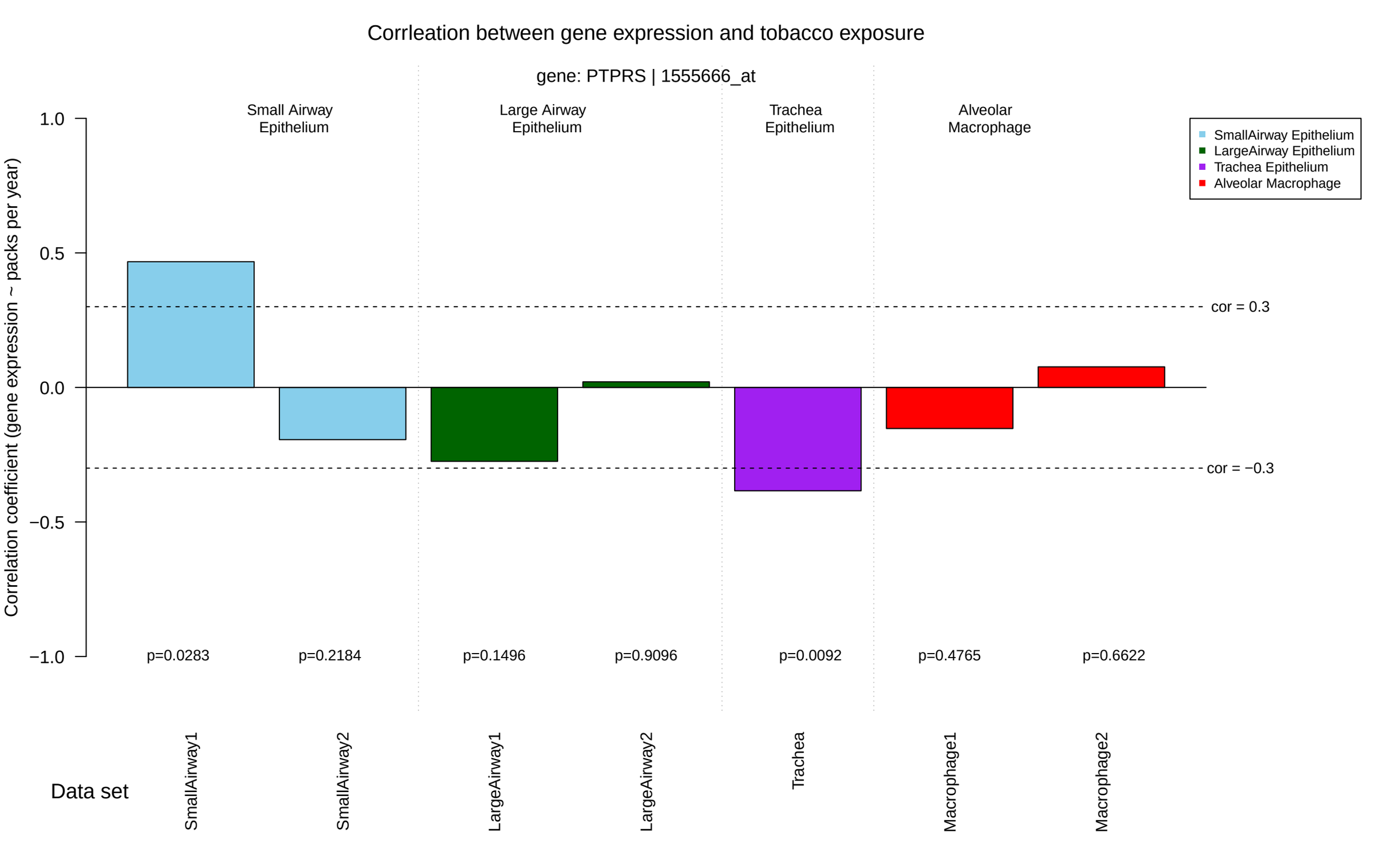

set: LargeAirway1 (p=0.7302)
PTPRS | 1555666_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9508)
PTPRS | 1555666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3074)
PTPRS | 1555666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5435)
PTPRS | 1555666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5504)
PTPRS | 1555666_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPRS | 1555666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0283
p=0.2184
p=0.1496
p=0.9096
p=0.0092
p=0.4765
p=0.6622
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2